

Guidelines for preparing your thesis

by Prof. Dr.-Ing. Lutz Leutelt, Dept. Informations- und Elektrotechnik, HAW Hamburg

not an official HAW document
not for the company
for use by my students only
still preliminary but hopefully useful

Dear student,

I am looking forward to taking over the supervision as your first examiner of your thesis. This document (which is still under development) shall give you some hints for a successful completion of your thesis.

Topic

A happy-end requires a good foundation. Hence, a good choice and design of the topic and the related tasks (work packages) of a Bachelor's (BA) or Master's (MA) thesis are crucial. For a Master's thesis the requirements are similar but higher regarding the aspects described in the following.

An appropriate topic gives you a **"big question"** for which you find step-by-step an answer in your thesis. Examples for "big questions" could be:

- "Is it possible to do X with Y?" (feasibility study),
- "How should an apparatus for Y look like that can do Z?" (design and development),
- "Which quality can be expected from X using Y?" (Evaluation).

Very often a topic includes aspects of all three questions and more.

A good topic gives you the chance to show your potential to find your own way and solution in an academic sense. Thus, it should not be too narrow that the solution is straightforward and that it should leave you enough room to explore, investigate, decide and implement something in your thesis. Particularly for a MA thesis, this is very important. In addition, it has higher requirements on complexity, theoretical depth and analysis of the state-of-the-art (literature study).

It is important that the topic is accurately described on 1-3 pages and that all sides agree with the topic: the university supervisor (who is me), the company supervisor (if thesis is done externally) and the student (who is you).

Sometimes academic goals (the interest of the university and the student) are contradicting to company goals. To give an example: the company wants to have a customer demonstrator for a trade show within 6 weeks for which a GUI is to be build that is already fully specified. This is a poor thesis topic since

- it puts you unnecessarily under time-pressure (you get 3 or 6 months from HAW Hamburg for your BA or MA thesis, respectively, not 6 weeks!),
- it doesn't leave you much to explore and decide as the GUI has already been fully specified and, finally,

- GUI designing has not the necessary depth and complexity required by an academic thesis.

So, hands-off such a topic! Your supervisor should/will not agree to a subject as described before.

BTW.: According to the examination regulations of the HAW Hamburg, it is the university that officially gives the topic to you, not the company!

And as a last remark: neither a BA thesis nor a MA thesis will be the scientific breakthrough in quantum physics or rocket sciences. It is not necessary that nobody before you did sth. in this field of science or R&D before. However, the topic should be unique given the particular boundary conditions and requirements. It must be clearly your own achievement, otherwise it is plagiarism if exactly the same was investigated before.

Timeline and Work Packages

The time scheduled for BA thesis and MA thesis is 3 months and 6 months, respectively. However, it is recommended that you take an additional **month** for **orientation** to get a feeling for the topic: How does it feel to work on this topic, with the new tools, with the supervisor, in that company? Can the “big question” probably be answered in the intended way? Is sufficient equipment, literature, knowhow available? Do supervisor and colleagues have time for you? After the orientation phase, the answers to these questions should be (more or less) positive.

It is possible that during this orientation phase the topic changes, because a **new focus for the “big question” was found**, the task turned out to be too complex that the topic needs to be narrowed, necessary equipment will not arrive on time or for many other reasons.

At least after the orientation phase, the **work packages** and **potential milestones** shall be defined and send to me. They should be detailed enough to give you orientation in the jungle of your thesis but not too detailed as small details can be hardly foreseen at the beginning. **Work packages of 3-10 working days are reasonable**. A good description of work packages is concise and precise: **What is to be done (activities)? What shall be the outcome (milestone)?** Typical work packages are

- **Orientation Phase**

Get an Understanding of: topic, underlying technology, processes and tools; maybe learn a new programming language, understand provided source code, study (in literature) what others did on the subject, ...

Outcome: you can explain the necessary fundamentals, judge whether the topic is a good one, know which steps need to be taken to accomplish the tasks

- **Requirement Analysis**

Which requirements are fixed (and why? By whom?) and which do you find yourself by further investigations on the “big question”? These can be customer requirements, technological limitations, resource limitations, usability aspects etc.

Outcome: structured and complete list of requirements

- **Concept Phase**

What are the parameters I can play with when finding the solution to the “big question”?

Which potential solutions are possible? Why is the one and only chosen concept for my solution the best with respect to the requirements? What do I need to test and evaluate to be sure?

Outcome: different concepts with assessment mapped to requirements

- **Design Phase** (Potentially you need to implement the concept by SW and/or HW), outcome: a program, an algorithm, a piece of hardware are a detailed specification of a system
- **Evaluation Phase**
Does the solution meet the requirements? In which way is it an answer to the “big question” and in which way not (yet)? What else need to be investigated if more time would be available to be fully sure?,
Outcome: performance assessment of your “product”

It depends on the topic which of the phases apply to your thesis. Sometime the focus is on evaluation, sometimes on the concept, sometimes on the design. But in the end the thesis will typically have all these aspects included in a way. The given order is typically also the temporal order. Do not make a concept, if requirements are not fully known. Do not implement, if the concept decision has not been well justified. And so on...

The very important last work package is the

- **Writing Phase**
Depending on the task, it consumes up to 30% of the total thesis time. Plan at least 4 weeks for a BA thesis and 6 weeks for a MA thesis. As a guideline, but not as a requirement, a BA thesis should have around 50-60 pages, a MA thesis 80-100 pages. You better should write precise and concise, than filling the thesis with less relevant material just for the sake of more pages.

Part-time thesis

Sometimes it is necessary that you work **part-time** on your thesis because you have to visit lectures in parallel or have to earn money with a job. In this case, the overall thesis time has to be scaled (“stretched”). The stretching factor depends on the amount of other activities you have in parallel. To give an example: if you work 2 day a week on your BA thesis, you have to stretch the overall time by a factor of 5/2, i.e. the time is $(1+3 \text{ months}) \cdot 5/2 = 10 \text{ months}$.

The part-time approach has the risk that you lose focus on your work and the progress is slowed down by constantly being interrupted in your workflow. I recommend to work not less than 3 days a week on the thesis, and those days preferably in a block (to avoid too many interruptions). During the writing phase you should consider focusing only on your thesis and work full-time on it if possible.

Academic working style

During your BA thesis it is probably the first time that you work independently over such a long time on such a complex topic. “Independently” means that you have to plan your activities yourself, make decisions on your own, have to motivate yourself again and again, even if you are in a difficult phase where the progress is smaller than expected (see section “motivation”).

Although you can and should (!) regularly request help from your supervisors throughout your thesis work, the help you can expect is often limited because, after a while, nobody else in the world will understand the details of the task as good as you do, the one who constantly works on it. So, you have consultants at your hand who can add different views and experiences, but in the end you have to find your way alone. Every day.

That requires both, a good overall work package and time planning and potentially readjustment over the entire thesis time, and a detailed planning of activities every week and day.

Once you started with your work, you will soon see that the number of results you collect and options you have will grow quickly and you will hold many unfinished threads in your hands. In order not to get lost, you should make sure that every step you take, every decision you make is made transparent and conserved for future reference by writing them down in a notebook or electronic text document. Otherwise, after few weeks, you will not remember many details anymore, which you have not written down. It is good practice to take 10-15 minutes every day to reflect the achievements, decision, problems and next steps by writing them down. There are tools like ePortfolios (see below) that can support you in this process and make the results easier accessible later.

If a major step was completed, e.g. the concept is finished, I recommended summarizing the major results in an electronic document with figures, graphs and calculations. It can be done in a draft way at that time: collect facts and conclusions just as bullet points.

This was about organizing and documenting your work. But what means “academic” work(ing style)? Basically that you work in a clean and traceable way. Every decision you make should be well explained and justified by applying the typically knowledge and skills of a last year student. For example, when you have to come up with a concept, you should discuss first what your options are, with which parameters you can “play” and then think in different directions about solutions (and write about them). The way you describe problem and solution shall use the formalisms and standards (expressions, terms, equations, notations, illustrations, naming conventions, abbreviations) common for the subject you study.

Any major decision you make or step you go shall be reflected with respect to the previously defined requirements. If you have several options, you should assess them (advantages, disadvantages, significance, impact). This approach shall be taken for all steps of your work, even for smaller decisions. For smaller decisions sometime half a sentence is sufficient for explanation, the bigger ones might need a detailed discussion.

Supervision

My role as a supervisor depends on whether your thesis is done externally or internally at university. If you do your thesis internally at HAW Hamburg, please read both following section as almost everything that is stated for external theses holds true for internal theses, too.

External thesis

When you write your thesis with a company, I expect that one or more members of staff closely supervise you. That incorporates that your supervisors have sufficient time and knowledge. If you come to the conclusion that this is not the case: make this explicit in the company and if nothing changes, change the topic (and company).

My role will be that I help you master the typical challenges of a thesis from an academic perspective. I will help you tune the topic until it is “good enough”, I will discuss the work packages with you and can be a discussion partner for major decisions if you need a different perspective on them. In case of trouble with your company (e.g. conflicting interests of company and academic goals), I try to moderate and bring in the (academic) university perspective.

If I do not hear anything from you, I assume that you are progressing and everything is fine. If you encounter problems that block your progress, you should immediately request a meeting by email with a description of the problem that you want to discuss.

“How am I writing?”

When it comes to writing, you must align the **structure of the thesis paper and the table of contents with me before you start the writing phase**. Once you wrote your first pages, send me **10 selected pages** on which you would like to have detailed feedback. Good selection candidates are pages from **introduction, requirement analysis** and/or **concept section**. You should even send the pages to me when you are not sure whether they are already “good enough”. You get my feedback “for free” and it has no impact on the later grade of your thesis.

The file format shall be pdf, in case you write with Latex, or docx, if you write with MS Word. Do not wait too long with sending me the ten pages that you can include my feedback as soon as possible to your further writing activities. Please note: depending on how busy I am at that moment, it might take up to 7-10 days until you get my feedback on your ten pages.

Internal thesis

Everything I wrote above on the external thesis is true for the internal thesis, too, except for the fact that I replace the company supervisor. In addition, I will help you find your way into the topic, discuss next steps and give technical hint where possible and necessary. But do not forget that soon you will be the expert for the task and nobody else will understand the details better than you. Instead of answers you might get a series of questions: “Did you already try this?”, “Can you find out whether ...?” to give you need threads to follow.

We should meet **at least every two weeks** and on demand more often. As mentioned above, you need to **trigger the meeting**, particularly **if you encounter problems** that block your progress. Let me know when you are unhappy with my style of supervision and have different expectations. It is important for the progress that we are a good team, each of us in our roles. You do not need to be afraid that such a feedback from you will have a negative impact on the grade. Actually, the opposite is true: when you are frustrated by supervision, motivation will drop off and endanger the success of your work.

Writing and Citing

There is a lot of literature available about scientific writing available – also in our HAW Hamburg library – that will describe scientific writing much better than I could do it. However, I would like stress following aspects.

Outline

The structure of the thesis basically follows the work pages described above:

Introduction – Fundamentals – Requirement analysis – Concepts – Implementation – Evaluation and Results – Summary and outlook

That does not mean that these terms already represent the headings of the sections. They depend on the topic and will be discussed before your start writing. There are different philosophies with which chapter to start writing. Some do it from the beginning to the end. Others write the chapters Requirements – Concepts – Implementation - Evaluation first and afterwards Introduction,

Fundamentals and Summary. The advantage of the latter is that you really know, what the final outcome your thesis will be and can directly guide the reader by Introduction and Fundamentals to the point, where he or she needs to be to understand your work.

Writing vs. editing

There is a fundamental difference between the writing phase and the editing phase:

“Writing is a proactive process: Whether one is given a topic or comes up with one, writing is an act of creation in which the writer calls forth the idea, the scope, the tone, and the structure of the work” [1]. This is the phase in which you collect and structure the information of the entire thesis and every section. You make sure that what you want to transfer is put on paper, in the right order, with a logically correct flow of thoughts. The focus is on completeness of information and the way you want to guide the reader through the jungle of information provided by you.

“Editing, by contrast, is reactive: One is assigned a piece of content, and one’s task is to refine the writer’s effort, helping him or her achieve the goal he or she was reaching for” [1]. The editor helps to make the text “nicer”: the layout (font, placement of pictures, how the equations are set), improve the formal language, check whether the flow of thought is logical and the intended message reaches the reader.

You are both, writer and editor of your text, in one person and therefore the risk can occur that you do not sufficiently separate the two phases: Oh, how tempting it is to tinker about the size, the position and orientation of figures while in the writing phase! Look at your watch, how much time that costs and how much attention it has been distracted from your writing activity. An the good thing is that it will have been for nothing, as you will add another paragraph to your text next day that you can start from scratch with formatting the figures.

So try to separate the writing and editing phase. Start collecting and compiling the information first and do not pay too much attention to layout and exact wording, This will be the focus in the editing phase. To find out whether a text is really written appropriately for the intended reader you need some distance (a few days) to the writing phase. It might be also useful to split every day in writing and editing phase. When you feel fresh in the morning: write! When a bit tired after lunch or in the evening: edit! The latter also includes drawing of figures and diagrams, reformatting of plots.

From general to specific

This rule is very important when composing both, your overall thesis paper as such and each of your major sections and paragraphs. You always start very general, introduce gently, with a broader scope, the idea you want to write about and then go step-by-step to the details and specifics of the problem.

Why is this good style and necessary?

You have become too much of an expert of the problem discussed in your thesis and the corresponding background. So much has become self-evident to you since the start of your thesis work and there are so many details that you want to share with the readers. So be aware that you do not overstrain your readers by well planning the “dose” and pace at which you give the information.

To find the right starting point, it is a good idea to assume that a fellow-student, who did not further look into your problem yet, is the intended reader of your thesis. With other words, if we could turn back time, it is you, yourself, who has to understand the thesis *before* you had started working on it for weeks and months. Before you write a chapter or section, think which information you would have needed first to understand the main outcome of the upcoming section.

By the way: the first sentences of the Introduction can be even more general. Maybe your parents and friends who are not about to become engineers would like to have an idea of what you did in your thesis, too.

And finally, the thesis paper will not be a complete image of all your findings and activities. Completeness is not the goal and you have to find the right balance between “too much” and “too little”. E.g. when you have a lot of data or source code or many plots, it is advisable to give an overview which data was generated, how and why in the main section of the thesis, but put the details into the appendix of the thesis paper or on CD-ROM.

References and Correct Citing

Citations are references to other documents that you included or considered in your thesis. The purpose of citation is manifold:

- You clearly separate your own work from the work others. Scientifically correct manner requires this as you must not claim the work of others as your own. Contempt of the rule would be plagiarism and will result in a fail. Even years after submission of your thesis. Another aspect is that you are not taking over liability for the correctness of foreign sources, i.e. if you cite figures from a publication that later turn out to be incorrect and/or manipulated, you can just be blamed for relying on these figures but not for the manipulation itself. Hence, citing also protects your own work.
- You make transparent on which foundation your investigations have been built. A well-researched reference list, well interwoven in the train of thought of your thesis shows that you have a broad overview of the field in which you work.
- You give the reader of your thesis the chance to research some of the aspects, which you could not describe in detail, at a greater depth.

There are many resources, which tell you how to correctly cite in a scientific publications, e.g. [2]. I gave you two examples ([1], [2]) how this could be done. For me, it is particularly important that your list of references only includes items for which a citation (number in square brackets “[78]”) in the text occurs. Latex takes over the major job for you and MS Word supports automatic citation, too.

When you extract information or ideas from an external source (document), but include them in your own words, it is sufficient to place the number in square brackets close to the included information. If it is an entire paragraph, you typically place the citation after the introduction sentences to the paragraph or at the end of the paragraph.

This, however is not sufficient, if you cite literally as done with [1] above. “Literally” mean that you do not just include the idea the external source but also a major part of the wording. Even if you leave out or change a few words of the external source, it is still a literal citation. In such a case, you have to put the cited text in quotes or italics, as shown above, and give the number in square brackets.

Large Language Models and Generative AI

HAW Hamburg has no specific guidelines for the use of AI tools in theses. You can find all current guidelines from the HAW here: <https://www.haw-hamburg.de/ki-in-der-lehre/> (German only, use a translator if necessary). In principle, the approval/prohibition of the use of AI tools is in the hands of the individual lecturers/the degree program or the department.

I personally allow the use of generative AI tools, but these places in your work must be clearly marked in the text. The added value and the result must be critically reflected and different tools must be used and compared with each other.

Motivation

Motivation stems from *movere* (Latin für move) and is the force that moves you forward during your thesis. There are many reasons, why this motivation can drop. Less progress than expected, too many threads and too little focus, too high expectations, family issues, mental problems and many more. The only way to overcome these problems is to talk. To friends, family and of course your supervisors. They can all help to find a way out of the motivation problem.

When the reasons are directly thesis related, the supervisors can help you find solutions to the seemingly overwhelming problem, adjust the focus of the topic, give you orientation by defining next steps together, revise the planning, give you an external perspective on your problem and progress. Believe it or not, even your supervisors did undergo (and still do from time to time) motivational problems and have an idea of how this feels. However, it is you to talk about your motivational problems first.

References

[1] Mark Nichol. (2017, August 18). "What's the Difference Between Writing and Editing?" DAILYWRITING TIPS [Online]. Available: <https://www.dailywritingtips.com/whats-the-difference-between-writing-and-editing/>

[2] Institute of Electrical and Electronics Engineers (IEEE). (2017, August 18). "IEEE Citation Reference" [Online]. Available: <http://www.ieee.org/documents/ieeecitationref.pdf>